

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Smart Study Generator Agent

Domain of Project –Education

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Problem Statement - Smart Study Generator Agent

The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students.

The solution should:

- 1. Fuse Learning Resources:** Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.
- 2. Enable Multimodal Input:** Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.
- 3. Create a Predictive & Agentic Layer:** Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., “Focus on Topic X—frequently asked in exams”).
- 4. Visualize Study Progress:** Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts. .

Technology Used

- **Langflow platform** -For designing and orchestrating multi-agent workflows visually.
- **IBM Granite model** - for eg – ibm-granite-4-h-small – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Intelligent Shopping Agent efficiently.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance.

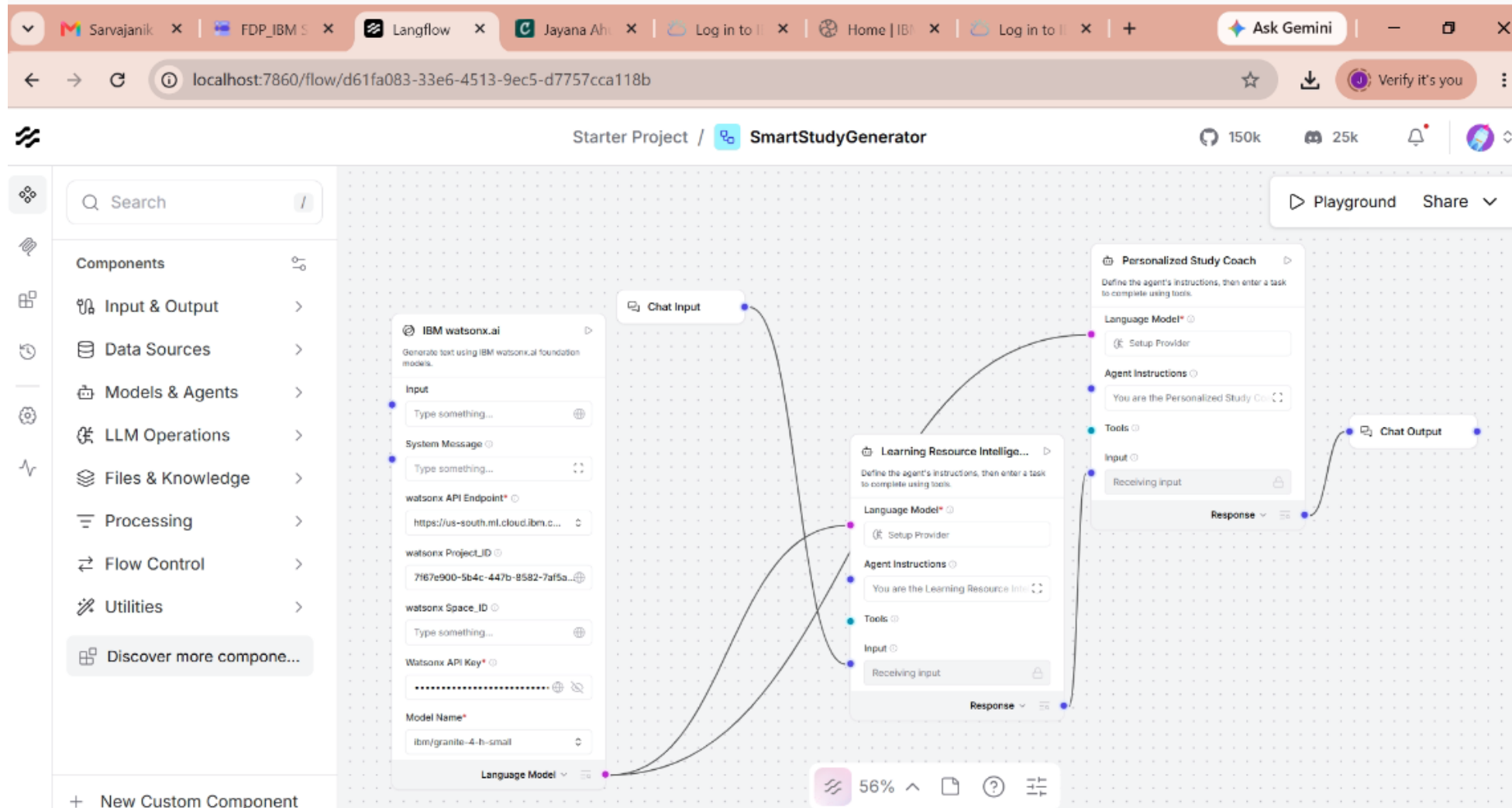
Proposed Solution

Develop an **Agentic AI-powered Smart Study Generator** using **IBM Granite Models**, **RAG**, and a **two-agent architecture**. The first agent collects, processes, and summarizes learning materials from multiple sources (notes, textbooks, images, voice, and online resources) into structured study content. The second agent analyzes student progress to create personalized study plans, generate adaptive quizzes, predict exam readiness, and recommend important topics. A real-time dashboard visualizes learning progress, weak areas, and performance forecasts, providing students with an intelligent, personalized, and adaptive learning experience.

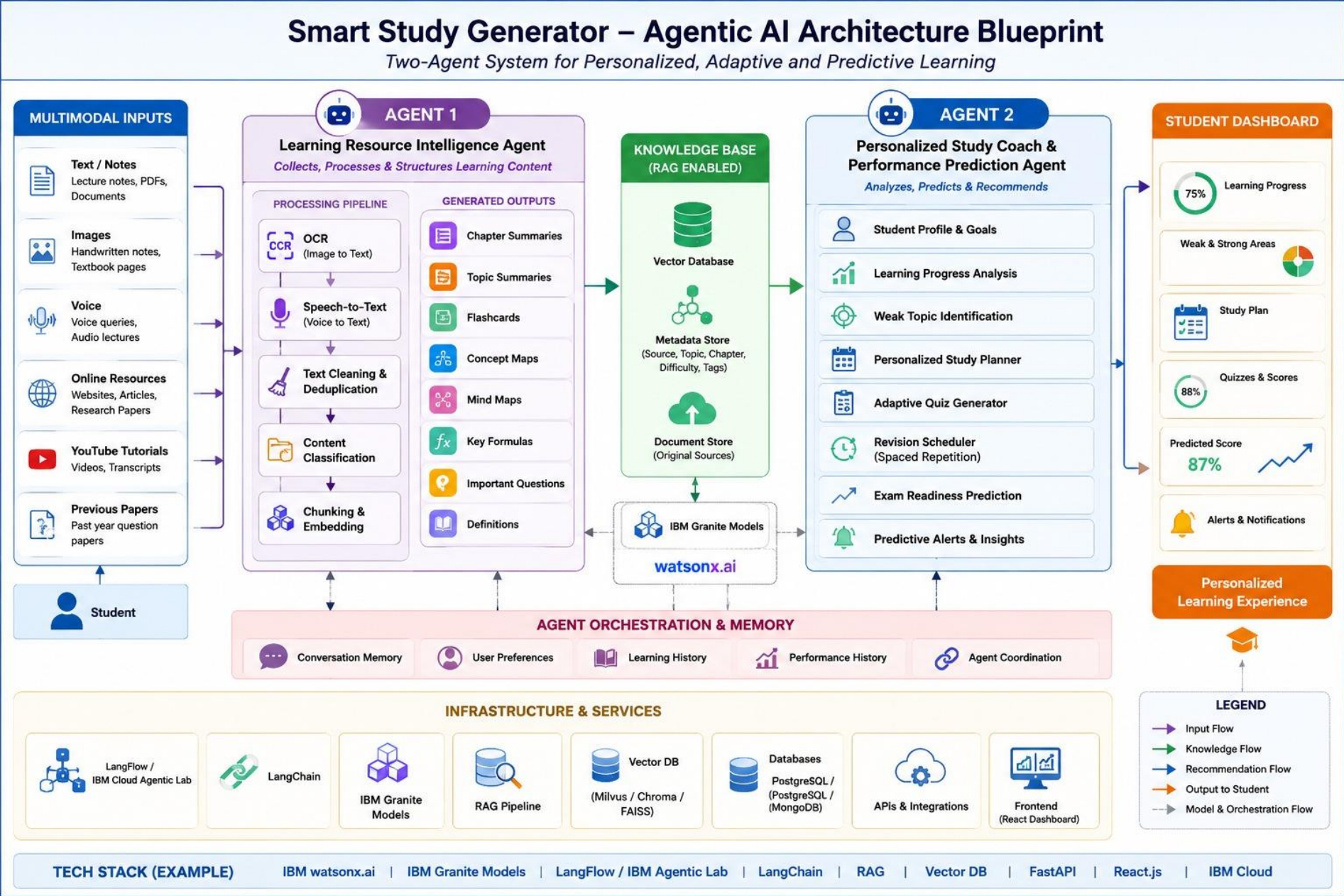
Langflow component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized shopping insights and recommendations.
- 3) **Name of IBM Granite model** - ibm-granite-4-h-small → Fast, efficient for real-time shopping recommendations.
- 4) **Chat output** -Displays AI-generated responses such as which material has to use

Langflow workflow -

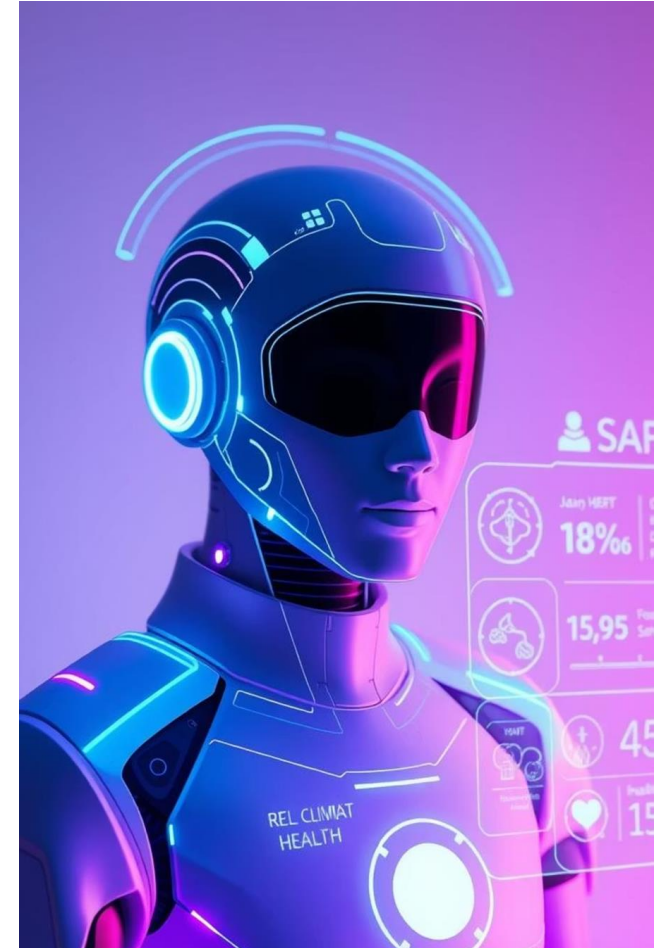


Architecture blueprint



Role of Agentic AI in the solution

Agentic AI enables autonomous collaboration between specialized AI agents to deliver a personalized learning experience. One agent organizes and transforms diverse study resources into structured content, while the other analyzes student performance to generate adaptive study plans, predictive recommendations, and real-time learning insights. This collaborative approach creates an intelligent, proactive, and continuously improving study companion.



Token Usage

10806 token are used

Before Prompt

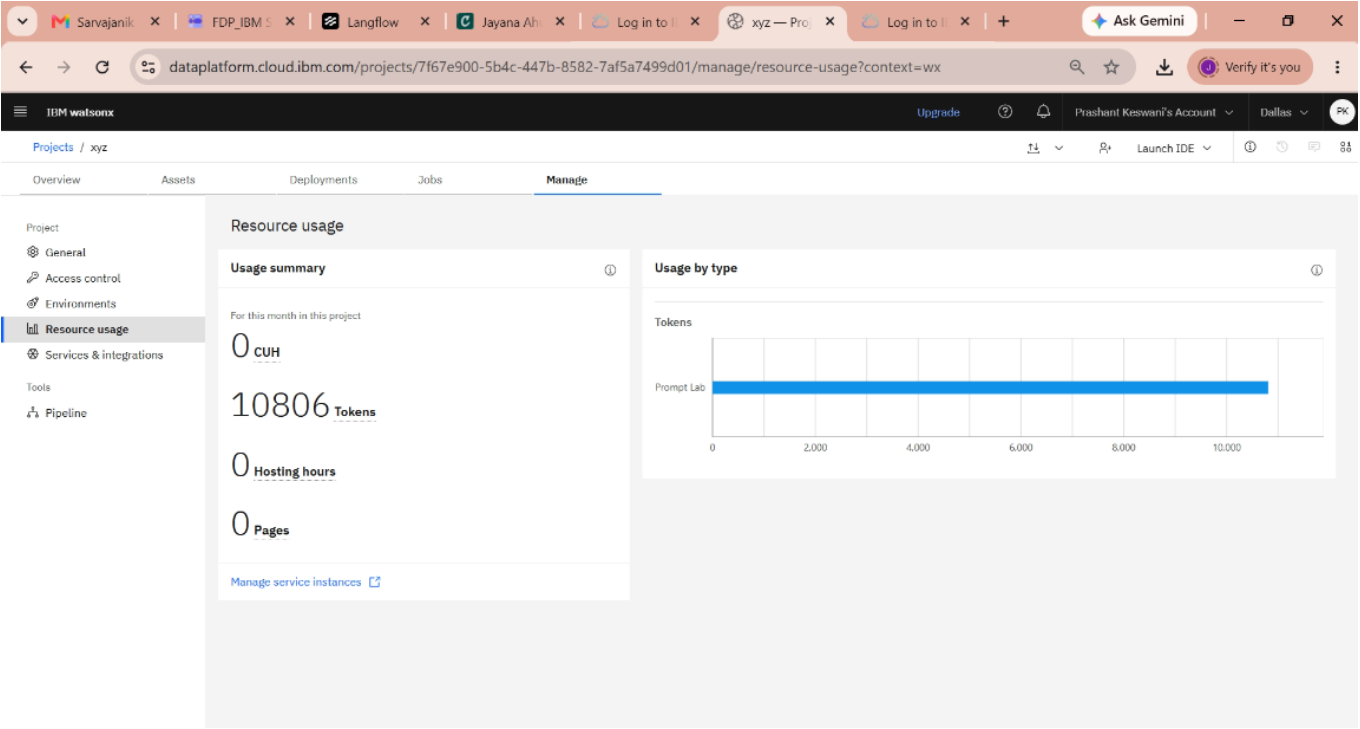
- Tokens : 3233
- Forgot to take screenshot before giving prompt

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After the Prompt



Output

Sarva

FDP_

Lang

Jayan

Log i

IBM v

Log i

Zoon

New

Ask Gemini

localhost:7860/flow/d61fa083-33e6-4513-9ec5-d7757cca118b

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Verify it's you

Starter Project /

SmartStudyGenerator

🗨 150k

💬 25k

🔔

Sessions

+

Default Session

⋮

☐ Select All

New Session 0

⋮

New Session 0

×

User

Create short question and answer on Introduction to Python

AI

✓ Finished in

🗨 4.0K | 25.8s

Introduction to Python – Short Questions & Answers

#	Question
1	What is Python and why is it popular for beginners?
2	How do you run a Python script from the command line?

🗨

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Novelty and Uniqueness

The Smart Study Generator combines **multimodal learning**, **RAG-based knowledge retrieval**, and a **two-agent Agentic AI architecture** to deliver personalized study materials, adaptive learning plans, predictive exam insights, and real-time progress tracking. Unlike traditional study tools, it continuously learns from student interactions and dynamically adapts to individual learning needs.

Git Hub Link

- GitHub Public repository link:

https://github.com/prashantkeswani-commits/FDP_IBM_Project_SmartStudyGeneratorAgent

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Certificate – Generative AI in Action

In recognition of the commitment to achieve
professional excellence



Prashant Keswani

Has successfully satisfied the requirements for:

Generative AI in Action



Issued on: Jun 23, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/266084cf-b688-4cf1-8966-57d7c939e984>



Certificate – Getting Started with Cybersecurity

In recognition of the commitment to achieve
professional excellence



Prashant Keswani

Has successfully satisfied the requirements for:

Getting Started with Cybersecurity



Issued on: Jun 23, 2026

Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/00bd0d20-3be8-480d-a716-92811a1906bb>



Certificate – Level Up Cybersecurity with Generative AI

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Prashant Keswani

for the completion of

Level Up Cybersecurity with Generative AI

(ALM-COURSE_3947352)

According to the Adobe Learning Manager system of record

Completion date: 25 Jun 2026 (GMT)

Learning hours: 1 hr 30 mins

Certificate – Exploring Quantum Computing

IBM SkillsBuild

Completion Certificate



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Exploring Quantum Computing

(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 25 Jun 2026 (GMT)

Learning hours: 50 mins

Certificate – Lab: Troubleshoot your code using IBM Bob

IBM **SkillsBuild**

Completion Certificate



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for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 30 mins

Future Scope

- Future Scope
- Integrate AR/VR-based immersive learning for interactive education.
- Support multiple languages to make learning accessible to a wider audience.
- Incorporate real-time collaboration for group study and peer learning.
- Enable integration with Learning Management Systems (LMS) and educational platforms.
- Enhance predictive analytics using advanced AI to provide personalized career and skill recommendations.
- Extend support to lifelong learning and professional certification programs.
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Thank You!

Thank you for your time and interest.